

Malaria Chemoprophylaxis

Patient Group Direction, version 007, issued 10 September 2026. Three arms: atovaquone/proguanil, doxycycline and mefloquine.

Scope of this PGD

This PGD authorises the supply of malaria chemoprophylaxis for travellers to malarious areas, following a destination risk assessment.

- **PROPHYLAXIS ONLY.** This PGD does not authorise the treatment of malaria in any circumstance. A traveller with fever, whether abroad or within a year of returning, is a medical emergency and must be referred the same day for a blood film.
- **WEIGHT DETERMINES THE DOSE AND THE PRODUCT** in children. Atovaquone/proguanil is supplied in two strengths and the paediatric strength must be used below 40kg. Weigh every child; do not estimate from age.
- The quantity must cover the pre-travel lead-in, the whole time in the malarious area, and the post-travel tail, which differs by agent. See Appendix 2.
- No chemoprophylaxis is completely effective. Bite avoidance is not optional advice and must be given and recorded in every case.
- Destination recommendations change. Use current NaTHNaC / TravelHealthPro country guidance for every consultation; do not work from memory.

Arm 1 , Atovaquone with proguanil

Summary of UKHSA ACMP guidance for malaria prevention

UKHSA, UK Malaria Expert Advisory Group (UKMEAG, formerly the Advisory Committee on Malaria Prevention), Guidelines for malaria prevention in travellers from the United Kingdom, 2026 (Chiodini PL, Patel D, Goodyer L, Chiodini J, Frise C, Mabayoje D, Aggarwal D; London: UKHSA, January 2026). Published on GOV.UK as HTML chapters 17 June 2026, page last updated 31 July 2026. Chapters read: Executive summary, General issues, Awareness of risk, Chemoprophylaxis, Diagnosis, Special risk groups. Summarised 10 September 2026.

This summary is part 2 of the house format for a Get Real Health PGD: what the PGD is for, then the guidance that governs the condition, then the PGD itself. It is here so that a pharmacist can hold this document against the guidance it claims to follow. It summarises the source named above and adds nothing to it.

The ABCD approach and how to give the advice

The guidelines are updated and reissued annually for healthcare workers who advise UK-based travellers to malaria-endemic areas, and may not be appropriate for people who live in endemic areas. UK health professionals are advised to use the UKMEAG guidelines as their preferred source, and the same recommendations appear on TravelHealthPro; both should be checked regularly because recommendations may change.

Malaria prevention advice is a combination of measures, the ABCD of malaria prevention: Awareness of risk, Bite prevention, Chemoprophylaxis, and Diagnose promptly and treat without delay.

The traveller should be told that while no regimen is 100 percent effective, the combination of preventive measures advised will give significant protection against the potentially severe consequences of malaria.

Based on individual risk assessment, discuss the choices of chemoprophylaxis regimen and their individual advantages and disadvantages, including cost. Use visual aids such as distribution maps, and show examples of bite prevention aids.

Provide the traveller with written information on malaria and its prevention. UKMEAG suggests that in every scenario a written record of the malaria prevention measures advised is given to the traveller so that they can pass it to their GP; a template for risk assessment and summary of advice is in Appendix 2.

As part of a stringent individual risk assessment a full clinical history is essential, detailing current medication (including hospital-prescribed drugs that may not appear on the GP repeat list), significant health problems and any known drug allergies.

Malaria prevention is only one aspect of pre-travel advice; an overall risk-assessment-based package of travel health advice should be provided.

How the choice of drug is made: destination, resistance and the traveller

Recommendations for antimalarials should be appropriate for the destination and tailored to the individual, taking into account possible risks and benefits to the traveller.

The choice of preventive measures should be based on the information in the Country recommendations table, supported by the text of the guidelines. The likelihood of malaria transmission may vary considerably within one country, and in-country maps are provided for selected countries.

Exposure depends on the number of infectious bites received. Risk is affected by temperature, altitude (parasite maturation usually cannot take place above 2,000 metres, though it has been reported up to 2,500 metres) and season; rural areas generally carry higher risk than urban, although in sub-Saharan Africa urban malaria is now highly prevalent; type of accommodation (an insecticide-impregnated bed net should be used unless the room has functioning air conditioning switched on and well-sealed doors and windows); patterns of activity, since *Anopheles* can bite at dawn, dusk, through the night and in some settings by day, indoors, in vehicles and at the airport on arrival; and length of stay, with longer stays carrying higher risk.

Chloroquine-resistant falciparum malaria is now widespread and effectively universal. Chloroquine-resistant *P. vivax* is found in the Indonesian archipelago, the Malay Peninsula including Myanmar, and eastward to Thailand, Cambodia and southern Vietnam, and is also reported from Ethiopia and South America. Significant *P. falciparum* resistance to mefloquine is a problem only in some areas of South-East Asia, with sporadic reports from the Amazon basin.

The drugs are not listed in order of preference; the preferred prophylaxis is determined by a full risk assessment for each individual traveller.

Causal prophylaxis (atovaquone plus proguanil) acts on the liver stage and is continued for 7 days after leaving a malarious area. Suppressive prophylaxis (mefloquine, doxycycline) acts on the red blood cell stage and is continued for 4 weeks after leaving. None of the prophylactic drugs advised acts against the dormant hypnozoite stage of *P. vivax* or *P. ovale*, so attacks of these can occur long after chemoprophylaxis ends; this is not drug failure. Routine use of primaquine is not recommended. Proguanil alone (Paludrine, and the Paludrine/Avloclor pack) has been discontinued in the UK. Avloclor and the atovaquone plus proguanil combination are not affected.

The SmPCs and BNF should be consulted for full prescribing information. Where a UKMEAG recommendation does not follow the SmPC, UKMEAG recommends following the UKMEAG recommendation.

Antimalarials purchased in the tropics may be fake or sub-standard, so travellers should obtain their medication from a reputable source in the UK before travel, and check that any internet supplier is bona fide.

Suspected side effects, and attacks of malaria in people prescribed prophylaxis, should be reported through the MHRA Yellow Card scheme; malaria cases should also be reported to the Malaria Reference Laboratory.

Atovaquone plus proguanil

Adult tablets contain atovaquone 250 mg and proguanil 100 mg; paediatric tablets contain atovaquone 62.5 mg and proguanil 25 mg. Taken orally with food or a milky drink. Diarrhoea or vomiting may reduce absorption of atovaquone.

Adult dose (40 kg or more): one tablet daily, starting 1 to 2 days before entering the malarious area, continuing throughout the stay and for 7 days after leaving. UKMEAG does not support abbreviated post-travel regimens. Prophylactic efficacy against *P. falciparum* is 90 percent or more.

Maloff Protect, a brand of the combination, can be purchased from UK pharmacies for adults aged over 18 weighing more than 40 kg.

Contraindications: allergy to proguanil, atovaquone or any tablet ingredient; renal impairment (avoid for prophylaxis if eGFR is less than 30 mL/minute/1.73 m²); not to be used in patients receiving renal dialysis.

Cautions: other agents are usually advised in pregnancy, but UKMEAG advises it may be used if there are no other appropriate options, with folic acid 5 mg daily (see special groups). It can be used when breastfeeding if there is no suitable alternative.

Interactions: the proguanil component may enhance the anticoagulant effect of warfarin, so the INR should be monitored. Plasma atovaquone is reduced by rifabutin, rifampicin, most non-nucleoside reverse transcriptase inhibitors, boosted HIV protease inhibitors, tetracycline and metoclopramide, with possible therapeutic failure, so concomitant use should be avoided. The antifolate effect of proguanil is increased with pyrimethamine. Check the HIV Drug Interactions website for antiretrovirals.

The most frequent side effects are headache and gastrointestinal upsets.

Children's doses by weight (Table 5): 5 to 7.9 kg, half a paediatric tablet daily; 8 to 9.9 kg, three-quarters of a paediatric tablet; 10 to 19.9 kg, one paediatric tablet; 20 to 29.9 kg, two paediatric tablets; 30 to 39.9 kg, three paediatric tablets; 40 kg and over, four paediatric tablets, better given as one adult tablet. Paediatric tablets are licensed from 11 kg upwards; the doses below 11 kg are UKMEAG recommendations. It can be given to infants weighing 5 kg or more.

Doxycycline

Capsules (50 or 100 mg) or dispersible 100 mg tablets. Swallow with plenty of fluid while sitting or standing and do not lie down for at least one hour afterwards, to reduce oesophageal irritation and ulceration. If gastric irritation occurs, take with food or milk; absorption is not notably affected by food or milk. Separate from antacids containing aluminium, calcium or magnesium and from oral zinc, iron or bismuth preparations as far as possible.

Dose 100 mg daily, starting 1 to 2 days before entering the malarious area, continuing throughout the stay and for 4 weeks after leaving. Efficacy is comparable to mefloquine.

Contraindications: allergy to tetracyclines or any ingredient; children under 12 years of age irrespective of weight (only licensed in the UK from 12 years because of potential bone damage and tooth discolouration). For children 25 to 44.9 kg the BNFC advises 100 mg once daily from 12 years.

Pregnancy: use is contraindicated in the SmPC and doxycycline is best avoided for prophylaxis in pregnancy. However, if required before 15 weeks' gestation it should not be withheld if other options are unsuitable; the whole course, including the 4 weeks after travel, must be completed before 15 weeks' gestation.

Breastfeeding: contraindicated in the SmPC and BNF. UKMEAG's view is that other options are preferable but doxycycline can be used if other options are unsuitable; courses longer than a month should be avoided wherever possible.

Cautions: hepatic impairment, patients taking potentially hepatotoxic drugs, renal impairment (though it may be used in severe renal failure), myasthenia gravis, systemic lupus erythematosus; candida infections may occur.

Photosensitivity: the prescriber should warn against excessive sun exposure and advise on correct use of a broad-spectrum sunscreen. Photosensitivity is mostly mild and transient. Doxycycline preparations have a low pH and may cause oesophagitis and gastritis, especially on an empty stomach or just before lying down, and as a broad-spectrum antibiotic may predispose to vaginal candidiasis.

Interactions: metabolism is accelerated by carbamazepine and phenytoin; try to advise another antimalarial, and if that is not possible or acceptable, increase to 100 mg twice daily and counsel on minimising adverse effects. Tetracyclines possibly enhance the anticoagulant effect of coumarins such as warfarin (the manufacturer advises INR monitoring) and doxycycline may increase ciclosporin levels. It is a non-enzyme-inducing antibiotic, so additional contraceptive precautions are not required with combined or progestogen-only oral contraceptives unless vomiting or diarrhoea occur. It possibly reduces the efficacy of oral typhoid vaccine and should preferably not be started within 3 days after the last vaccine dose.

Doxycycline is the simplest chemoprophylaxis for most people taking antiretrovirals, but the SmPC, BNF and the traveller's HIV physician should be consulted.

Mefloquine

250 mg tablets, taken orally, preferably after a meal and with plenty of liquid. Stringent risk assessment is required before advising mefloquine.

Adult dose (45 kg or more): 250 mg weekly, starting 2 to 3 weeks before entering the malarious area to assess tolerability, continuing throughout the stay and for 4 weeks after leaving. Protective efficacy is 90 percent or more.

Contraindications: allergy to mefloquine or any tablet ingredient; allergy to quinine or quinidine; a current or previous history of depression, generalised anxiety disorder, psychosis, schizophrenia, suicide attempts, suicidal thoughts, self-endangering behaviour or any other psychiatric disorder, or epilepsy or convulsions of any origin; a history of Blackwater fever; severe impairment of liver function; use with halofantrine (and halofantrine should not be given within 15 weeks of the last mefloquine dose); and use with Zyban (bupropion SR) because the chance of seizure may be increased.

The risk of epilepsy and serious mental health disorders is higher in first-degree relatives of people with these conditions, so a family history should be sought; a condition in a first-degree relative may not contraindicate an antimalarial but may influence the choice. A checklist should be used to ensure proper screening before mefloquine is given, and the manufacturer's SmPC, patient leaflet and risk materials should be consulted.

Cautions: pregnancy and breastfeeding (see special groups), cardiac conduction disorders; not recommended in infants under 5 kg. The SmPC states that periodic liver function checks and eye assessments should be performed if used for a prolonged period, and anyone on mefloquine who presents with a visual disorder should be referred to their treating physician. After traumatic brain injury the decision is made individually after a detailed risk assessment.

Mefloquine lowers the seizure threshold and its side effects could be confused with decompression or narcosis events, so it may be better avoided for diving holidays; some sub-aqua centres do not permit divers taking it. The UK Civil Aviation Authority advises it should not be given to pilots.

Side effects: neuropsychiatric problems and vestibular disorders. Those taking mefloquine are more likely to have abnormal dreams, insomnia, anxiety and depressed mood during travel than those taking atovaquone-proguanil or doxycycline, with increased neuropsychiatric adverse events especially in women. Dizziness, balance disorder, tinnitus and vertigo may occur and in a small number of patients may continue for months after stopping the drug.

The psychiatric symptoms to be regarded as prodromal for a more severe event are insomnia, abnormal dreams or nightmares, acute anxiety, depression, restlessness and confusion. Overall mefloquine remains an important agent, tolerated by most travellers who take it.

Children's doses by weight (Table 3): 5 to 9.9 kg, a quarter tablet (the SmPC allows use above 5 kg); 10 to 15.9 kg, a quarter tablet (0.375 would be preferable but cannot be safely provided by breaking the adult tablet); 16 to 24.9 kg, half a tablet; 25 to 44.9 kg, three-quarters of a tablet; 45 kg and over, the adult dose of one tablet. There is no suspension, so adult tablets must be broken.

Renal disease: may be used in severe renal impairment and no dose reduction is needed in dialysis. Liver disease: contraindicated in severe hepatic impairment (SmPC), can be used in mild or moderate impairment after discussion with the patient's specialist.

Special groups

Pregnancy: pregnant women are advised to avoid travel to malarious areas. They have an increased risk of severe malaria and a higher risk of fatality. If travel is unavoidable they should be told the risks of malaria and the risks and benefits of chemoprophylaxis, and bite avoidance is extremely important: ideally remain indoors between dusk and dawn, and use DEET in a concentration of not more than 50 percent.

Table 8 in pregnancy: atovaquone plus proguanil is not routinely advised due to sparse data, but if there are no other appropriate options it may be used in all trimesters with folic acid 5 mg daily co-prescribed if conception is planned and in all trimesters, for as long as the drug is taken, because of a theoretical increased risk of neural tube defects from the antifolate action. Chloroquine is safe in all trimesters. Doxycycline should be avoided, but before 15 weeks' gestation should not be withheld if other options are unsuitable, and the course including the 4 weeks after travel must be completed before 15 weeks. Mefloquine: caution in the first trimester but can be used in all trimesters for travellers to high-risk areas; where transmission and resistance make it first choice, it may be used in the second and third trimesters, and its use is also justified in the first trimester in areas of high falciparum risk such as sub-Saharan Africa.

Women who have taken mefloquine or atovaquone-proguanil inadvertently just before or during the first trimester should be advised that this is not grounds to terminate the pregnancy. Travellers planning to conceive after prophylaxis who wish minimal drug present may observe these intervals after finishing the course: mefloquine 3 months, doxycycline 1 week, atovaquone-proguanil 2 weeks. Seek expert advice if a woman plans to conceive during a visit to a high-risk chloroquine-resistant area.

Breastfeeding: mothers should take the usual adult dose appropriate to the destination. The amount in breast milk will not protect the infant, so the breastfed child needs their own prophylaxis. Table 9: chloroquine safe; mefloquine, experience suggests safe; doxycycline and atovaquone-proguanil not recommended but can be used if there is no suitable alternative. Breastfeeding women should wash repellent off their hands and breast skin before handling infants.

Children: children with malaria may deteriorate very rapidly, so carers, family and school staff should know that a child who becomes unwell within a year of leaving a malarious area needs medical attention and a malaria blood test without delay. Parents are advised against taking infants and young children to malarious areas without adequate precautions. Weight is a better guide than age, so children should be weighed for dose calculation; parents should supervise dosing and not exceed maximum doses, since antimalarials can be particularly toxic to children. In the absence of syrups, mefloquine or atovaquone-proguanil tablets may be broken and crushed if necessary and mixed with condensed milk, butter, oily spreads, full-fat yoghurt, jam, honey or similar to mask the bitter taste. When quarter or half tablets are needed, the pieces from the same tablet should be given on consecutive days to minimise dose variation.

Epilepsy: a history of febrile convulsions only does not contraindicate any of the drugs. Table 11: atovaquone-proguanil can be used; chloroquine is contraindicated; doxycycline can be used but avoid with phenytoin, carbamazepine and barbiturates because its half-life may be reduced, and if unavoidable increase to 100 mg twice daily; mefloquine is contraindicated.

Renal impairment: dose reduction is required only in severe renal impairment. Atovaquone-proguanil is not recommended if eGFR is less than 30 mL/minute and not to be used in dialysis; doxycycline may be used in severe renal impairment; mefloquine may be used in severe renal impairment with no dose reduction in dialysis.

Liver disease: choice should be made after discussion with the patient's specialist. Mefloquine is contraindicated in severe impairment; doxycycline can be used with caution in severe impairment and the dose does not have to be adjusted; the atovaquone-proguanil manufacturer anticipates no special precautions or dose adjustment in severe impairment.

Warfarin: travellers should inform their anticoagulant clinic and start their malaria tablets 2 to 3 weeks before departure, with a baseline INR before starting and a re-check after one week, and again once prophylaxis is completed. Mefloquine inhibits CYP3A4 and P-glycoprotein and could increase DOAC concentrations; seek haematology advice if doubt remains.

G6PD deficiency: atovaquone-proguanil, doxycycline and mefloquine can all be used. Sickle cell disease: rigorous protection is essential. Splenectomy or severe hyposplenism: particular risk of severe malaria; avoid travel where possible, and if unwell during or after travel seek medical attention urgently.

Elderly travellers: no dose reduction is required on the basis of age alone, but they are more likely to have renal impairment or take other medication, which may influence choice; they are at particular risk of dying from malaria once acquired.

Emergency standby treatment in outline

Emergency standby treatment should be recommended for those taking chemoprophylaxis and visiting remote areas where they are unlikely to be within 24 hours of medical attention. It is for travellers who believe they may have malaria and is not a replacement for chemoprophylaxis.

It should be started if it is impossible to consult a doctor and/or reach a diagnosis within 24 hours of the onset of fever, and medical attention should still be sought as soon as possible to exclude other serious causes of fever.

The traveller should complete the standby course and recommence their chemoprophylaxis one week after the first treatment dose. If quinine is used, mefloquine should be resumed at least 12 hours after the last quinine dose. A second full dose should be taken if vomiting occurs within 30 minutes (half dose if after 30 to 60 minutes). Antipyretics should be used for fever.

The standby agent should be different from the chemoprophylaxis drug, both to minimise toxicity and because of resistance concerns. Written instructions are required, covering symptoms suggesting malaria including fever of 38 degrees C and above, when to start, how to take it, expected side effects and the possibility of drug failure; a traveller leaflet is in Appendix 3.

Adult regimens (Table 7): artemether plus lumefantrine 20/120 mg, 4 tablets initially then 4 tablets at 8, 24, 36, 48 and 60 hours (24 tablets over 60 hours) taken with food; or atovaquone plus proguanil 250/100 mg, 4 tablets as a single dose on each of 3 consecutive days; or quinine 300 mg 2 tablets three times a day for 3 days with doxycycline 100 mg twice daily for 7 days. In pregnancy artemether-lumefantrine is preferred in all trimesters, with quinine plus clindamycin as the alternative. Sulfadoxine-pyrimethamine and halofantrine are not recommended, and dihydroartemisinin-piperaquine cannot currently be recommended for this indication. Standby medication should also be obtained from a reputable UK source.

What to tell the traveller

Explain the ABCD approach, and that no regimen is 100 percent effective but the combination of measures gives significant protection.

Explain exactly when to start, that the tablets are taken throughout the stay, and for how long after leaving (7 days for atovaquone-proguanil, 4 weeks for doxycycline and mefloquine); the full course must be completed.

Give the drug-specific advice above: sun protection and upright posture with doxycycline; the prodromal psychiatric symptoms that mean stopping mefloquine and seeking advice; taking atovaquone-proguanil with food or a milky drink.

Suspected malaria is a medical emergency. Fever on return from the tropics should be considered to be malaria until proven otherwise, and malaria should be considered in every ill patient who has returned from the tropics in the previous year, especially in the previous 3 months. Symptoms are non-specific: fever, sweats or chills, malaise, muscle pain, diarrhoea, cough. Malaria cannot be diagnosed or excluded on clinical features, so an urgent blood film is needed; there is no need to wait for a fever spike, and one negative film does not exclude the diagnosis.

Vivax or ovale malaria can present long after prophylaxis has finished because of the hypnozoite stage; this is not a failure of the tablets.

Obtain antimalarials from a reputable source in the UK before travel; antimalarials bought in the tropics may be fake. Keep all medicines out of the sight and reach of children; chloroquine overdose in particular can be fatal.

Report suspected side effects via the Yellow Card scheme, and give the traveller a written record of the advice for their GP.

Patient Group Direction for the supply of atovaquone/proguanil for malaria chemoprophylaxis in adults and children weighing 11kg and over.

TWO STRENGTHS. Malarone Paediatric 62.5mg/25mg is used from 11kg to 40kg. The adult 250mg/100mg tablet is used only above 40kg. Supplying the adult tablet to a child in the 11 to 20kg band gives four times the intended atovaquone dose.

Healthcare professionals covered, and training

- Pharmacist registered and practising with the GPhC.
- Pharmacy technician registered and practising with the GPhC. None of these medicines is a controlled drug, so registered pharmacy technicians may lawfully supply under this PGD (Human Medicines Regulations 2012 as amended, 26 June 2024).
- Must have completed training relevant to travel health, documented and overseen by the Get Real Health team.
- Must be able to obtain and record an accurate BODY WEIGHT for every child, and to apply the weight bands in Appendix 1. Weight, not age, determines the dose and the presentation.
- Must be competent in destination risk assessment using current NaTHNaC / TravelHealthPro country recommendations, and must not select an agent without doing so.
- Must be able to calculate the full course quantity including the pre-travel lead-in and the post-travel tail, using Appendix 2.
- Must previously have used PGDs to supply medication.
- Must work in compliance with the SOPs of their own employer and practise only within the bounds of their own competence.

The PGD

Indication	Malaria chemoprophylaxis for travel to an area where atovaquone/proguanil is recommended, in adults and children weighing 11kg and over.
Inclusion criteria	• Weight 11kg or more, measured and recorded. • Travelling to an area where chemoprophylaxis is recommended, established from current NaTHNaC / TravelHealthPro guidance. • Able to take a daily oral dose for the whole course including 7 days after leaving the area. • Valid informed consent given, from a person with parental responsibility where the patient is a child. • No exclusion criterion present.
Exclusion criteria	• No destination risk assessment carried out, or the destination does not require chemoprophylaxis. • Any febrile illness at the time of presentation, or a fever at any time within the last 12 months following travel to a malarious area that has not been investigated with a malaria blood film. Malaria is a medical emergency; refer the same day. • Treatment of suspected or confirmed malaria. This PGD covers PROPHYLAXIS ONLY. A febrile returning traveller needs urgent hospital assessment and a blood film, not a prophylactic dose. • Pregnancy or breastfeeding. Refer; chemoprophylaxis in pregnancy needs individual specialist assessment and is outside this PGD. • Weight below the minimum for every available agent, or weight not obtainable. • Unable to give valid informed consent, or unwilling to complete the full course including the post-travel tail. • Weight below 11kg. • Known hypersensitivity to atovaquone or proguanil. • Known severe renal impairment, or the patient reports kidney disease or dialysis. Refer. • Taking rifampicin, rifabutin, metoclopramide or tetracycline, which reduce atovaquone concentrations. Refer. • Taking antiretroviral therapy (efavirenz, other NNRTIs or boosted protease inhibitors), or pyrimethamine. Refer. • Taking warfarin. Refer for INR monitoring rather than supplying.

Cautions	• Take with food or a milky drink; absorption is materially reduced on an empty stomach and this affects protection. • If a dose is vomited within an hour, repeat it. • Advise that this agent is started only 1 to 2 days before travel, so a traveller presenting late can still be covered.
Actions if excluded or declines	Explain why this agent cannot be supplied and whether another arm of this PGD is suitable. Where fever is present or has occurred since travel, refer the same day for urgent assessment and say plainly that malaria must be excluded. Where no agent is suitable, refer to a travel clinic or the GP, and give bite avoidance advice regardless. Document the advice and the decision.

The medicine

Name, form and strength	Atovaquone 250mg with proguanil hydrochloride 100mg tablets (adult strength). Atovaquone 62.5mg with proguanil hydrochloride 25mg tablets (paediatric strength). Both strengths must be stocked to run this arm for children. Confirm which strength is in your hand before supplying.
Legal category	POM.
Route and method	Oral, once daily at the same time each day, with food or a milky drink.
Dose and frequency	11 to 20kg: ONE paediatric tablet (62.5/25mg) once daily. 21 to 30kg: TWO paediatric tablets once daily. 31 to 40kg: THREE paediatric tablets once daily. Over 40kg: ONE adult tablet (250/100mg) once daily. Start 1 to 2 days before entering the malarious area, continue daily throughout, and continue for 7 days after leaving.
Quantity to be supplied	Calculate as: 2 days lead-in + days in the area + 7 days tail. Multiply by the tablets per day for the weight band. Worked example, adult, 14 day trip: 2 + 14 + 7 = 23 days = 23 adult tablets. Worked example, 25kg child, 14 day trip: 23 days at 2 paediatric tablets = 46 paediatric tablets. See Appendix 2. Supply the whole course. A short supply leaves the traveller unprotected at the end, which is when risk is highest.
Maximum treatment period	Continuous use for up to 12 months is supported. Beyond that, refer for specialist advice.
Adverse effects	Common: abdominal pain, nausea, vomiting, diarrhoea, headache. Uncommon: mouth ulcers, rash, raised liver enzymes. Rare: severe skin reactions.
Yellow Card reporting	Report suspected adverse reactions via https://yellowcard.mhra.gov.uk and inform the GP.
Records to be kept	• That valid informed consent was given, and from whom where the patient is a child. • Patient name, address, date of birth, and the GP with whom they are registered. • BODY WEIGHT in kilograms for every patient under 40kg, and the weight band applied. • Destination, itinerary, departure and return dates, and the source consulted for the destination recommendation. • Which agent was chosen and why, including why any alternative was unsuitable.

	• Name, form and STRENGTH of the product supplied. Malarone and Malarone Paediatric are different strengths and both must be recorded by name. • Date of supply, dose, quantity supplied, and the calculated course length including the tail. • Batch number and expiry date. • Advice given, including bite avoidance and the post-travel fever warning. • Details of any adverse drug reactions and the actions taken. • That the medicine was supplied under this PGD. Records signed, dated, legible and contemporaneous. Adults: 8 years. Under 18s: until the 25th birthday.
Storage	Store below 25C in the original packaging.

Information for the patient

Written information	Supply the patient information leaflet for the strength given, and the Get Real Health bite avoidance and post-travel fever sheet.
Counselling	• Take every day at the same time, with food or a milky drink. Missing doses or taking them on an empty stomach reduces protection. • Keep taking it for 7 days after you leave the malaria area. Stopping when you get home is the commonest reason prophylaxis fails. • No tablet is completely effective. Avoid bites: repellent containing 20 to 30% DEET or 20% picaridin, cover up at dusk and after dark, and sleep under an insecticide-treated net. • If you develop a fever at any time up to a YEAR after returning, seek medical help immediately and tell them you have been to a malaria area. Malaria can kill within 24 hours. • If you are sick within an hour of a dose, take another one.
Follow-up	No routine follow-up. Seek urgent medical attention for any fever during or after travel, and say where you have been.

Arm 2 , Doxycycline

Patient Group Direction for the supply of doxycycline 100mg for malaria chemoprophylaxis in patients aged 12 years and over.

Healthcare professionals covered, and training

- Pharmacist registered and practising with the GPhC.
- Pharmacy technician registered and practising with the GPhC. None of these medicines is a controlled drug, so registered pharmacy technicians may lawfully supply under this PGD (Human Medicines Regulations 2012 as amended, 26 June 2024).
- Must have completed training relevant to travel health, documented and overseen by the Get Real Health team.
- Must be able to obtain and record an accurate BODY WEIGHT for every child, and to apply the weight bands in Appendix 1. Weight, not age, determines the dose and the presentation.
- Must be competent in destination risk assessment using current NaTHNaC / TravelHealthPro country recommendations, and must not select an agent without doing so.
- Must be able to calculate the full course quantity including the pre-travel lead-in and the post-travel tail, using Appendix 2.
- Must previously have used PGDs to supply medication.
- Must work in compliance with the SOPs of their own employer and practise only within the bounds of their own competence.

The PGD

Indication	Malaria chemoprophylaxis for travel to an area where doxycycline is recommended, in patients aged 12 years and over.
Inclusion criteria	• Aged 12 years and over. • Travelling to an area where chemoprophylaxis is recommended, established from current NaTHNaC / TravelHealthPro guidance. • Able to take a daily dose for the whole course including 4 weeks after leaving the area. • Valid informed consent given. • No exclusion criterion present.
Exclusion criteria	• No destination risk assessment carried out, or the destination does not require chemoprophylaxis. • Any febrile illness at the time of presentation, or a fever at any time within the last 12 months following travel to a malarious area that has not been investigated with a malaria blood film. Malaria is a medical emergency; refer the same day. • Treatment of suspected or confirmed malaria. This PGD covers PROPHYLAXIS ONLY. A febrile returning traveller needs urgent hospital assessment and a blood film, not a prophylactic dose. • Pregnancy or breastfeeding. Refer; chemoprophylaxis in pregnancy needs individual specialist assessment and is outside this PGD. • Weight below the minimum for every available agent, or weight not obtainable. • Unable to give valid informed consent, or unwilling to complete the full course including the post-travel tail. • Under 12 years of age. Doxycycline causes permanent tooth discoloration and enamel hypoplasia in younger children. • Known hypersensitivity to doxycycline or other tetracyclines. • Known severe hepatic impairment. • Taking warfarin. Doxycycline potentiates it; refer for INR monitoring. • Taking carbamazepine, phenytoin, phenobarbital or rifampicin, which reduce doxycycline concentrations. Refer. • Taking isotretinoin. Refer. • Systemic lupus erythematosus, myasthenia gravis or porphyria.
Cautions	• Photosensitivity is common and matters in a sunny destination.

	Advise high-factor sunscreen, covering up and avoiding midday sun. • Swallow with plenty of water, sitting or standing, and not immediately before lying down, to avoid oesophageal ulceration. • Separate from antacids, iron and dairy by at least 2 hours. • May increase the risk of vaginal candidiasis. Warn women and consider advising them to carry treatment.
Actions if excluded or declines	As Arm 1.

The medicine

Name, form and strength	Doxycycline 100mg capsules.
Legal category	POM.
Route and method	Oral, once daily, swallowed with plenty of water sitting or standing, and well before lying down.
Dose and frequency	100mg once daily. Start 1 to 2 days before entering the malarious area, continue daily throughout, and continue for 4 WEEKS after leaving.
Quantity to be supplied	Calculate as: 2 days lead-in + days in the area + 28 days tail. Worked example, 14 day trip: 2 + 14 + 28 = 44 capsules. The minimum possible supply for any trip is therefore about 31 capsules. A quantity below 30 cannot be correct for any itinerary.
Maximum treatment period	Continuous use for up to 2 years is supported. Beyond that, refer.
Adverse effects	Common: nausea, diarrhoea, photosensitivity, vaginal candidiasis. Uncommon: oesophageal irritation or ulceration, rash, headache. Rare: benign intracranial hypertension, hepatotoxicity, severe skin reactions.
Yellow Card reporting	Report suspected adverse reactions via https://yellowcard.mhra.gov.uk and inform the GP.
Records to be kept	• That valid informed consent was given, and from whom where the patient is a child. • Patient name, address, date of birth, and the GP with whom they are registered. • BODY WEIGHT in kilograms for every patient under 40kg, and the weight band applied. • Destination, itinerary, departure and return dates, and the source consulted for the destination recommendation. • Which agent was chosen and why, including why any alternative was unsuitable. • Name, form and STRENGTH of the product supplied. Malarone and Malarone Paediatric are different strengths and both must be recorded by name. • Date of supply, dose, quantity supplied, and the calculated course length including the tail. • Batch number and expiry date. • Advice given, including bite avoidance and the post-travel fever warning. • Details of any adverse drug reactions and the actions taken. • That the medicine was supplied under this PGD. Records signed, dated, legible and contemporaneous. Adults: 8 years. Under 18s: until the 25th birthday.

Storage	Store below 25C in the original packaging.
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Information for the patient

Written information	Supply the doxycycline patient information leaflet, and the Get Real Health bite avoidance and post-travel fever sheet.
Counselling	● Keep taking it for FOUR WEEKS after you leave the malaria area. This is much longer than some other tablets and it is not optional. ● Take with plenty of water, sitting or standing up, and not just before bed. ● You will burn much more easily in the sun. Use high-factor sunscreen and cover up. ● Avoid indigestion remedies, iron tablets and milk within 2 hours of a dose. ● No tablet is completely effective. Avoid bites as described on your sheet. ● If you develop a fever at any time up to a YEAR after returning, seek help immediately and say where you have been.
Follow-up	As Arm 1.

Arm 3 , Mefloquine

Patient Group Direction for the supply of mefloquine 250mg for malaria chemoprophylaxis in adults and children weighing 5kg and over.

Mefloquine is a weekly agent started 2 to 3 weeks before travel, specifically so that tolerability can be assessed before departure. Do not supply it to a traveller leaving within 2 weeks.

Healthcare professionals covered, and training

- Pharmacist registered and practising with the GPhC.
- Pharmacy technician registered and practising with the GPhC. None of these medicines is a controlled drug, so registered pharmacy technicians may lawfully supply under this PGD (Human Medicines Regulations 2012 as amended, 26 June 2024).
- Must have completed training relevant to travel health, documented and overseen by the Get Real Health team.
- Must be able to obtain and record an accurate BODY WEIGHT for every child, and to apply the weight bands in Appendix 1. Weight, not age, determines the dose and the presentation.
- Must be competent in destination risk assessment using current NaTHNaC / TravelHealthPro country recommendations, and must not select an agent without doing so.
- Must be able to calculate the full course quantity including the pre-travel lead-in and the post-travel tail, using Appendix 2.
- Must previously have used PGDs to supply medication.
- Must work in compliance with the SOPs of their own employer and practise only within the bounds of their own competence.

The PGD

Indication	Malaria chemoprophylaxis for travel to an area where mefloquine is recommended, in adults and children weighing 5kg and over, where there is sufficient time before travel to assess tolerability.
Inclusion criteria	• Weight 5kg or more, measured and recorded. • At least 2 to 3 weeks before departure, so that the first doses can be taken and tolerated before travel. • Travelling to an area where mefloquine is recommended, established from current NaTHNaC / TravelHealthPro guidance. • Valid informed consent given, from a person with parental responsibility where the patient is a child. • No exclusion criterion present.
Exclusion criteria	• No destination risk assessment carried out, or the destination does not require chemoprophylaxis. • Any febrile illness at the time of presentation, or a fever at any time within the last 12 months following travel to a malarious area that has not been investigated with a malaria blood film. Malaria is a medical emergency; refer the same day. • Treatment of suspected or confirmed malaria. This PGD covers PROPHYLAXIS ONLY. A febrile returning traveller needs urgent hospital assessment and a blood film, not a prophylactic dose. • Pregnancy or breastfeeding. Refer; chemoprophylaxis in pregnancy needs individual specialist assessment and is outside this PGD. • Weight below the minimum for every available agent, or weight not obtainable. • Unable to give valid informed consent, or unwilling to complete the full course including the post-travel tail. • Weight below 5kg. • ANY current or previous psychiatric disorder, including depression, anxiety, psychosis, or a suicide attempt at any time. • Epilepsy or any seizure disorder, or taking an anticonvulsant. • Taking bupropion, or any other medicine that lowers the seizure threshold. Refer. • Taking, or planning to take, halofantrine or ketoconazole.

	● History of Blackwater fever. ● Known hypersensitivity to mefloquine, quinine or quinidine. ● Known cardiac conduction disorder, or a family history of one. ● Severe hepatic impairment. ● Departing within 2 weeks, so that tolerability cannot be assessed before travel. ● Occupation requiring fine coordination or spatial discrimination, such as pilots and divers. Refer. ● Cardiac conduction disorders, including known QT prolongation, or a family history of either, and concurrent QT-prolonging medicines. Refer. Mefloquine carries QT prolongation and arrhythmia in its SPC as rare but serious.
Cautions	● Advise the patient to stop and seek advice if they develop anxiety, depression, restlessness, confusion or unusual behaviour. Neuropsychiatric effects may precede a more serious reaction. ● Take with food and plenty of water, on the same day each week. ● Prior good tolerance of mefloquine is reassuring, not a reason to exclude.
Actions if excluded or declines	As Arm 1. Where mefloquine is excluded on psychiatric or neurological grounds, note that this is an absolute exclusion and offer an alternative arm rather than seeking a second opinion.

The medicine

Name, form and strength	Mefloquine 250mg tablets.
Legal category	POM.
Route and method	Oral, once weekly on the same day each week, with food and plenty of water.
Dose and frequency	Over 45kg: ONE tablet (250mg) once weekly. 31 to 45kg: THREE QUARTERS of a tablet once weekly. 21 to 30kg: HALF a tablet once weekly. 5 to 20kg: ONE QUARTER of a tablet once weekly. Tablets must be scored to divide accurately. Confirm the product held is scored before supplying a divided dose; if it is not, refer. Start 2 to 3 weeks before travel, continue weekly throughout, and continue for 4 WEEKS after leaving.
Quantity to be supplied	Calculate as: 3 weeks lead-in + weeks in the area + 4 weeks tail, in weekly doses. Worked example, adult, 14 day trip: 3 + 2 + 4 = 9 weekly doses = 9 tablets. The minimum possible supply for any trip is therefore about 8 tablets.
Maximum treatment period	Continuous use for up to 1 year is supported. Beyond that, refer.
Adverse effects	Common: nausea, abdominal pain, diarrhoea, headache, dizziness, sleep disturbance and vivid dreams. Uncommon: anxiety, depression, restlessness, confusion. Rare but serious: psychosis, seizures, suicidal ideation, prolonged vestibular disturbance.
Yellow Card reporting	Report suspected adverse reactions via https://yellowcard.mhra.gov.uk and inform the GP. Neuropsychiatric reactions in particular should be reported.
Records to be kept	● That valid informed consent was given, and from whom where the patient is a child. ● Patient name, address, date of birth, and the GP with whom they are

	registered. ● BODY WEIGHT in kilograms for every patient under 40kg, and the weight band applied. ● Destination, itinerary, departure and return dates, and the source consulted for the destination recommendation. ● Which agent was chosen and why, including why any alternative was unsuitable. ● Name, form and STRENGTH of the product supplied. Malarone and Malarone Paediatric are different strengths and both must be recorded by name. ● Date of supply, dose, quantity supplied, and the calculated course length including the tail. ● Batch number and expiry date. ● Advice given, including bite avoidance and the post-travel fever warning. ● Details of any adverse drug reactions and the actions taken. ● That the medicine was supplied under this PGD. Records signed, dated, legible and contemporaneous. Adults: 8 years. Under 18s: until the 25th birthday.
Storage	Store below 25C in the original packaging.

Information for the patient

Written information	Supply the mefloquine patient information leaflet, the manufacturer alert card where provided, and the Get Real Health bite avoidance and post-travel fever sheet.
Counselling	● Take it on the SAME DAY each week, with food and plenty of water. ● STOP TAKING IT and seek advice if you become anxious, low in mood, restless, confused, or feel not yourself. Do not wait to see if it settles. ● Vivid dreams and disturbed sleep are common and are not by themselves a reason to stop, but tell us if they are distressing. ● Keep taking it for FOUR WEEKS after you leave the malaria area. ● No tablet is completely effective. Avoid bites as described on your sheet. ● If you develop a fever at any time up to a YEAR after returning, seek help immediately and say where you have been.
Follow-up	Contact the pharmacy if any neuropsychiatric symptom develops before travel, so an alternative can be arranged while there is still time.

Appendix 1 , Atovaquone/proguanil weight bands

Weight, not age, determines the dose AND the product strength. Weigh every child.

11 to 20 kg	ONE Malarone Paediatric tablet (62.5mg/25mg) once daily.
21 to 30 kg	TWO Malarone Paediatric tablets once daily.
31 to 40 kg	THREE Malarone Paediatric tablets once daily.
Over 40 kg	ONE adult tablet (250mg/100mg) once daily.
Under 11 kg	Outside this PGD. Refer.

Appendix 2 , Course length and quantity

Atovaquone/proguanil	- Start 1 to 2 days before entering the area. - Continue 7 DAYS after leaving. - Quantity = (2 + days in area + 7) x tablets per day for the weight band.
Doxycycline	- Start 1 to 2 days before entering the area. - Continue 4 WEEKS after leaving. - Quantity = 2 + days in area + 28 capsules.
Mefloquine	- Start 2 to 3 weeks before travel. - Continue 4 WEEKS after leaving. - Quantity = 3 + weeks in area + 4 weekly doses.

Supply the whole calculated course. The post-travel tail is the part travellers stop early and it is the part that prevents relapse and late presentation.

Appendix 3 , Bite avoidance, to be given in every case

- Insect repellent containing 20 to 30% DEET, or 20% picaridin, applied to exposed skin and reapplied as directed.
- Cover arms and legs from dusk onwards. The Anopheles mosquito bites between dusk and dawn.
- Sleep under an insecticide-treated bed net where accommodation is not screened or air-conditioned.
- Apply sunscreen first, then repellent.

Appendix 4 , The post-travel fever warning

Give this verbally and in writing to every patient.

Any fever occurring during travel, or at any time up to a year after returning from a malarious area, must be assessed urgently and the travel history disclosed. Falciparum malaria can be fatal within 24 hours of the first symptom. No chemoprophylaxis is completely effective, and a traveller who took their tablets correctly can still develop malaria.

Authorisation

Version	v007
Supersedes	Version 003, 9 September 2026
Valid from date	9 September 2026
Expiry date	31 July 2027

Doctor	Name: Nitin Shori Job title: Medical Director, Get Real Health GMC: 6047293 Signed: N. Shori Date: 9 September 2026
Pharmacist	Name: Chris Pilkington Job title: Head Pharmacist, Get Real Health GPhC: 2046322 Signed: C. Pilkington Date: 9 September 2026

Both authorising signatories reviewed this version together on 9 September 2026 and gave their authorisation for it to be issued. Signatures were applied digitally on their joint instruction, which is the established process for Get Real Health PGDs.

Adoption by the employing organisation

To be completed by the adopting pharmacy. These signatories are the adopting organisation own signatories and must not be pre-filled by Get Real Health.

Superintendent / clinical lead	Name: Job title and organisation: Signature: Date:
Professional group signatory	Name: Job title and organisation: Signature: Date:

Change history

Version	v004
Date	7 September 2026
Changes	• The practitioner Agreement to practise page, the premises block and the practitioner signature table are restored. Every version 001 document carried them; no document produced by the current generator did, through one omission in one function, which left 15 live documents with no page for an individual practitioner to sign. Raised as blocking by an adopting pharmacy: NICE MPG2 expects a record of the individuals authorised to work under a PGD. The standard governance requirements are restored at the same time: SPC and BNF familiarity, MHRA safety alerts, CPD and appraisal, indemnity, and capacity and consent, all of which were in version 001 and were lost in the rewrite. This version changes no clinical content. • The cardiac conduction and QT prolongation exclusion is restored to the MEFLOQUINE arm. Version 001 excluded cardiac conduction disorders including QT prolongation and a family history of either; the version 002 rewrite lost it, while mefloquine carries QT prolongation and arrhythmia as rare but serious effects in its SPC. It is restored to that arm only, not to the atovaquone with proguanil or doxycycline arms, where it does not apply. Found by comparing against the archived version 001. • The practitioner Agreement to practise page, the premises block and the practitioner signature table are restored. Every version 001 document carried them; no document produced by the current generator did, through one omission in one function, which left 15 live documents with no page for an individual practitioner to sign. Raised as blocking by an adopting pharmacy: NICE MPG2 expects a record of the individuals authorised to work under a PGD. The standard governance requirements are restored at the same time: SPC and BNF familiarity, MHRA safety alerts, CPD and appraisal, indemnity, and capacity and consent, all of which were in version 001 and were lost in the rewrite. This version changes no clinical content. • Full clinical review and reissue.

Standard requirements for every practitioner working under this PGD

All users must be familiar with the current Summary of Product Characteristics for every product named in this PGD, and with current BNF and national guidance for the condition.

All users must be up to date with MHRA drug safety alerts and recalls relevant to these products.

All users must be up to date with their CPD requirements and appraisal.

All users must hold appropriate professional indemnity covering this service.

All users must understand capacity and consent, including the Mental Capacity Act 2005, and must be able to assess consent in children and young people where this PGD covers them.

Agreement to practise by the registered healthcare professional

By signing below you confirm that you agree to the contents of this PGD and that you will work within it. A PGD does not remove the inherent professional obligations or accountability of the individual practitioner. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of the pharmacy or clinic premises to which this PGD relates

Premises name
 Address
 Postcode:
 ODS code, if applicable

Registered healthcare professionals authorised to work under this PGD at those premises

The employing organisation must keep this page, or an equivalent local register, and must be able to produce it on request. A practitioner who has not signed against the current version is not authorised to work under it.

[illegible]

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 Postcode:
 ODS code, if applicable

The employing organisation must keep this page, or an equivalent local register, and must be able to produce it on request. A practitioner who has not signed against the current version is not authorised to work under it.

[illegible]

Version and change record

Version: v005

Issued: 10 September 2026

Supersedes: Version 004, 9 September 2026

Change: the summary of the governing guidance is added as part 2 of the document, between the cover and the PGD, which is the house format for every Get Real Health PGD. It is summarised from the source it names and adds nothing to it. Nothing else in this document is altered, and this is not a clinical review of the remainder.

Version and change record

Version: v006

Issued: 10 September 2026

Supersedes: Version 005, 10 September 2026

Valid from: 10 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Validity: this version is valid from 10 September 2026 and expires on 31 July 2027. The signed master previously stated no period of validity, or an expiry of 31 October 2026; those are superseded.

Atovaquone/proguanil arm: antiretroviral therapy (efavirenz, other NNRTIs, boosted protease inhibitors) and pyrimethamine added to the exclusions. The SmPC says these should be avoided and the guidance summary named them; the exclusion did not.

Mefloquine arm: bupropion and other seizure-threshold-lowering medicines, halofantrine, ketoconazole and a history of Blackwater fever added as exclusions (SmPC 4.3 and 4.4); quinidine added to the hypersensitivity exclusion.

The weight bands (SmPC versus UKMEAG) are unchanged pending a decision and remain in the review register.

Version and change record

Version: v007

Issued: 10 September 2026

Supersedes: Version 006, 10 September 2026

Valid from: 10 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.

Changes in this version:

Fever exclusion window: one month in the exclusion criteria, twelve months on the cover and in every counselling box. Now 12 months everywhere: any fever following travel to a malarious area in the last 12 months that has not been investigated with a malaria blood film excludes and refers.

Signed on behalf of Get Real Health

Version v007, 10 September 2026.

Name	Qualification	Signature	Date
Nitin Shori	Medical Director GMC: 6047293		10 September 2026
Chris Pilkington	Head Pharmacist GPhC: 2046322		10 September 2026

Both authorising signatories reviewed this version together on 10 September 2026 and gave their authorisation for it to be issued. Signatures were applied digitally on their joint instruction, which is the established process for Get Real Health PGDs. This version is not valid without both signatures.